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Leaf anatomy of *Pinus thunbergii* Parl. (Pinaceae) collected from different regions of Korea

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곰솔의 잎 해부 형태

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ABSTRACT: Leaf anatomical study of *Pinus thunbergii* collected from 12 different coastal regions of Korea was conducted to understand the adaptive variation on leaf traits. Basic anatomical features are typical pine needle type with fibrous epidermis, 2-3 layered hypodermis, sunken stomata, monomorphic mesophyll, and well-represented bundle sheath. The bundle sheath surrounds a couple of vascular bundle separated by parenchyma bands. On the basis of their position, the resin ducts are of three types; external, medial and internal of the bundle sheath. The total number of resin ducts in all samples varies from 4 to 12. The stomata were found on stomatal bands throughout the leaf surface. Important dissimilarities observed on *P. thunburgii* leaf are the number and position of resin ducts and the number of stomata rows in leaf surface.

Keywords: *Pinus thunbergii*, leaf anatomy, resin ducts, stomata

적 요: 곰솔의 지역간 잎 형태 변이를 이해하기 위하여 한반도의 해안지역 12군데에서 채집한 곰솔의 잎 해부형태를 비교 연구하였다. 곰솔 잎의 해부학적 특징은 섬유상 표피, 2-3층의 하피, 표피보다 내려앉은 기공구조, 한가지 형태로 구성된 엽육조직, 뚜렷한 유관속초 등을 가진 전형적인 바늘상 잎이다. 곰솔 잎에 있어서 수지도는 발생위치에 따라 유관속초의 외부, 중간, 내부 등 3가지 분포형태로 구분되었다. 조사된 모든 재료에서 수지도의 개수는 4개에서 12개까지 지역간 변이가 관찰되었다. 기공은 잎 전체를 통하여 기공선으로 발달하였다. 곰솔 잎 해부형태에서 관찰된 중요한 차이는 수지도의 발생위치와 전체개수 그리고 잎 표면에 나타나는 기공의 기공선 개수이었다.

주요어: 곰솔, 잎 해부, 수지도, 기공

Pinus thunbergii Parl. is an evergreen, small, irregularly-shaped costal pine. This plant is one of the most important tree species in coastal forest of Japan and in other similar regions such as the shoreline in Shandong Province, China (Han et al., 2009; Murai et al., 1992; Zhang et al., 2011). The species also grows naturally in Korea and is one of the major constituents of the coastal forests established on sand dunes and ocean facing mountains (Obase et al., 2011). Tree trunks are usually not straight but sweep up in a gentle curve. *Pinus thunbergii* is the main species of windbreak forests along the coast of Korea including Ullung-do and Jeju-do. Usually it acquires the warm temperature to grow. The species has been thought to be stronger than other trees in coastal environment,

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and as a result, about 95% of the total coast is covered with it (Kim, 2005; Ministry of Environment, 2001). In Korea, the species is often misunderstood to be the native species of the coastal dune, probably because they have been traditionally planted in large scale since World War II (Choi et al., 2013).

Leaf features are often affected by environmental factors, as they are directly exposed to the environment. Variation in leaf structure among tree in different environmental conditions may be response to differences in light availability, temperature and moisture regimes in the habitat where they grow (Abrams and Kubiske, 1990). Some morphological differences shown by leaf trait could provide new information for determining the phylogenetic relationship of the species. Environmental factors show a direct response to stomatal pattern and position, number and size of resin ducts in *Pinus* species (Tiwari, 2013; Sheue et al. 2003). It has been said that morphological and anatomical features of *Pinus* needles hugely depend on abiotic factors (Fahn and Bemayoun, 1976; Schoettle and Rochelle, 2000). In vascular plants, the stomata are important for photosynthesis and transpiration. Therefore, the development of stomata is considered a key event in evolution of advanced land plants (Hetherington and Woodward, 2003). The differences in arrangement of stomata complex have a potential taxonomic value and thus can be useful in the selection of a taxonomic key to distinguish between Iberian Mediterranean pines (Garcia et al., 2009).

Some studies on *P. thunbergii* at coastal sites in Korea have been done, however none of these are related to taxonomy of the species (Choi et al., 2013; Kim, 2005; Kim et al., 2013; Obase et al., 2011; Yang et al., 2009). The objective of this study is to know the morphological differences on the leaf morphology of *P. thunbergii* collected from different coastal sites of Korea.

Materials and methods

Leaf materials of *P. thunbergii* were collected from 12 different coastal areas of Korea (Fig. 1). Twelve locations are basically categorized into five geographical regions: four sites belong to eastern coast, four sites to southern coast, two sites to western coast, Ullung-do and Jeju-do. Leaf materials were selected from middle parts of second year leaves. The number of population collected was shown in Table 1. The collected leaf materials were fixed in FAA (5 parts formalin; 5 parts glacial acetic acid; 90 parts 50% ethanol) and then preserved in 50% ethanol. Well matured healthy leaves were selected and cut into small pieces of about 5-6 mm. Three to four pieces per leaf each from middle and base were chosen and dehydrated through ethanol series (50%, 70%, 80%, 90%, 95%, 99% and absolute alcohol). After complete dehydration, the leaf materials were passed through

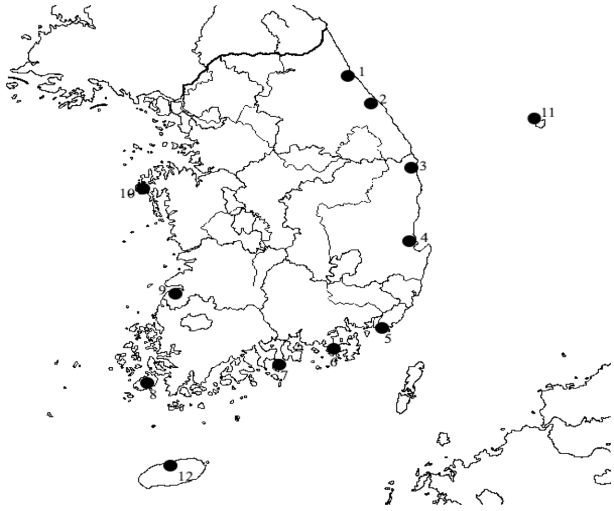


Fig. 1. The collection sites on the coast of Korea.

Sites	Annual precipitation (mm)	Mean temperature (°C)
Gangneung	1245.2	13.3
Busan	1591.1	14.7
Gochang-gusipo	1317.3	13.1
Ullungdo	1383.4	12.4
Jejudo	1497.5	12.9

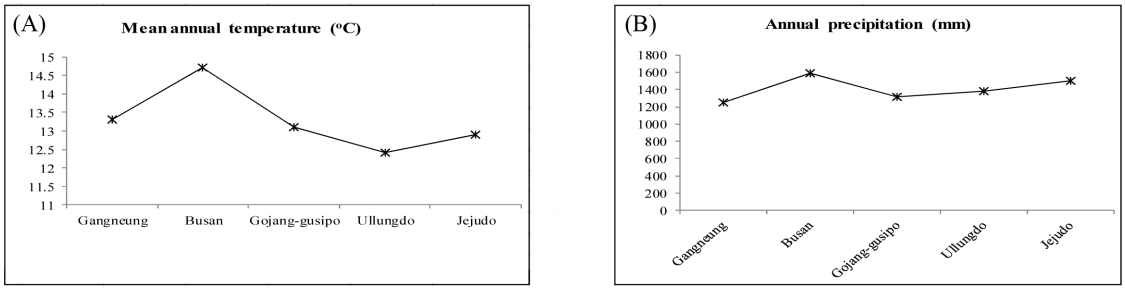


Fig. 2. Climatograms of the five *Pinus thunbergii* regions included in this study. **A.** Mean annual temperature, **B.** Annual precipitation.

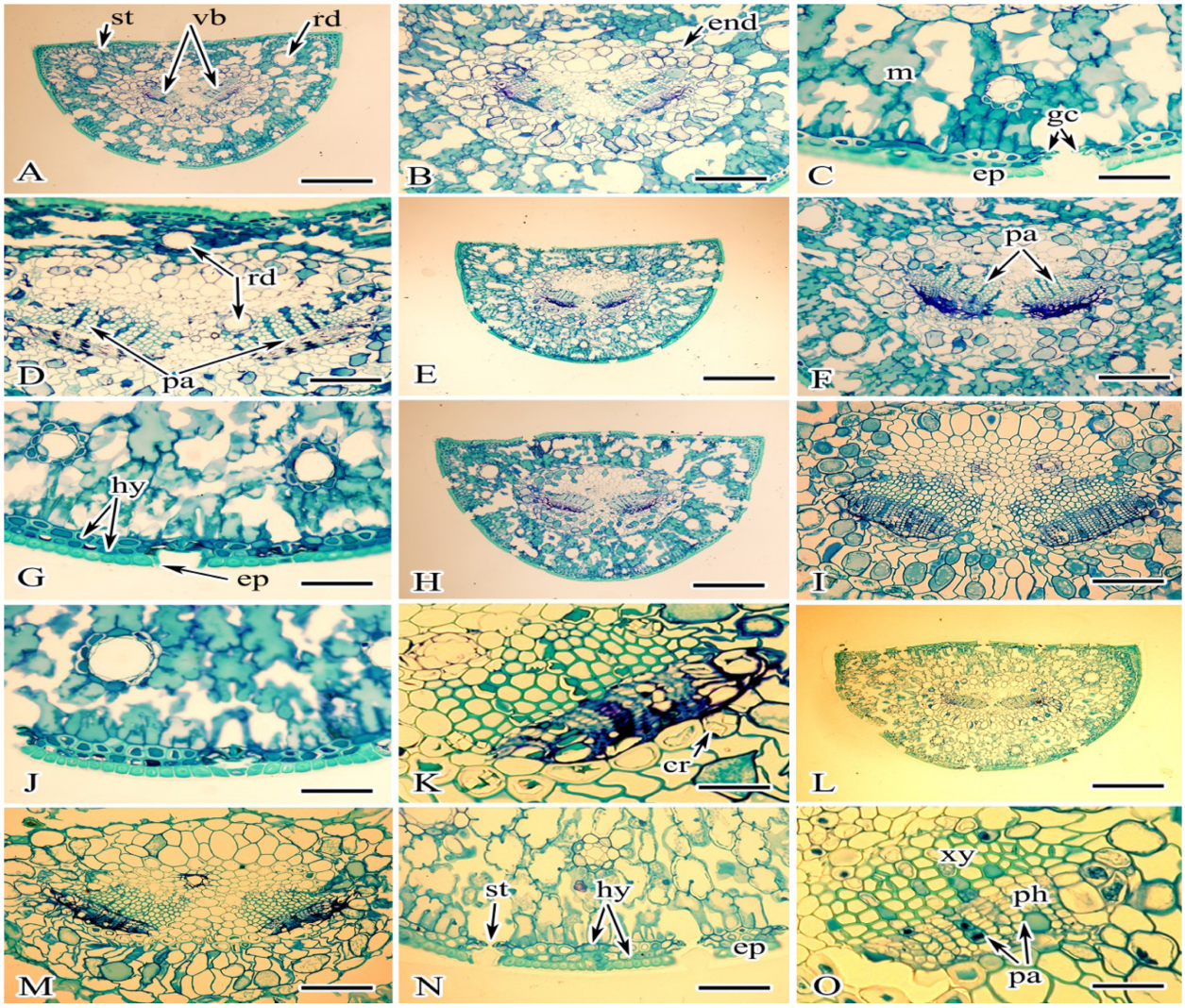


Fig. 3. Cross section (CS) of *Pinus thunbergii* leaf. **A-D.** Samples of Gangneung. **A.** CS of whole leaf. **B.** Vascular bundle. **C.** Abaxial epidermis showing stomata, epidermis and mesophyll. **D.** Adaxial epidermis and external resin duct. **E-G.** Samples of Pohang. **E.** Whole leaf. **F.** Vascular bundle showing parenchyma bands. **G.** Abaxial epidermis. **H-K.** Samples of Busan and Weinarodo. **H.** Whole leaf. **I.** Vascular bundle. **J.** Abaxial epidermis. **K.** Single vascular bundle showing crystals. **L-O.** Samples of Gochang-gusipo. **L.** Whole leaf. **M.** Vascular bundle. **N.** Abaxial epidermis. **O.** Single vascular bundle showing parenchyma bands. Abbreviations: cr, crystals; ep, epidermis; gc, guard cells; hy, hypodermis; m, mesophyll; pa, parenchyma bands; ph, phloem; rd, resin duct; st, stomata; vb, vascular bundle; vbs, vascular bundle sheath; xy, xylem. Scale bars: A, E, H, L = 75 μm; B, C, D, F, G, I, J, M, N = 20 μm; K, O = 10 μm.

alcohol + Technovit combinations (3:1, 1:1, 1:3, and 100% Technovit) and then embedded in Technovit 7100 resin. Embedded materials were then sectioned. Serial sections of 4-5 μm thickness were cut by disposable blades, stick on the slide glass and dried on electric warmer for 12 hours.

Dried slides were stained with 0.1% Toludine blue O for 60 to 90 seconds, rinsed with running water and again dried on electric slide warmer for more than six hours to remove water. The stained slides were then mounted with Entellan (Merck Co. Germany) and pressed with metal blocks for two days to remove air bubbles. After two days, permanent slides were prepared and observed under BX-50 light microscope (Olympus Co., Japan). Photographs were taken by digital camera system attached to microscope. For scanning electron microscopy, mature leaves were cut into small pieces and dehydrated through ethanol series same as in microtome method. After dehydration, they were dried with critical point dryer (Hitachi II, Hitachi Co., Japan), and coated with platinum by ion-sputter (E-1010, Hitachi Co., Japan). The microscopic pictures were taken under 15 kv by scanning electron microscope (S-3500N, Hitachi Co., Japan). Climate data were obtained from Korea meteorological administration for recent five years (2009~2013).

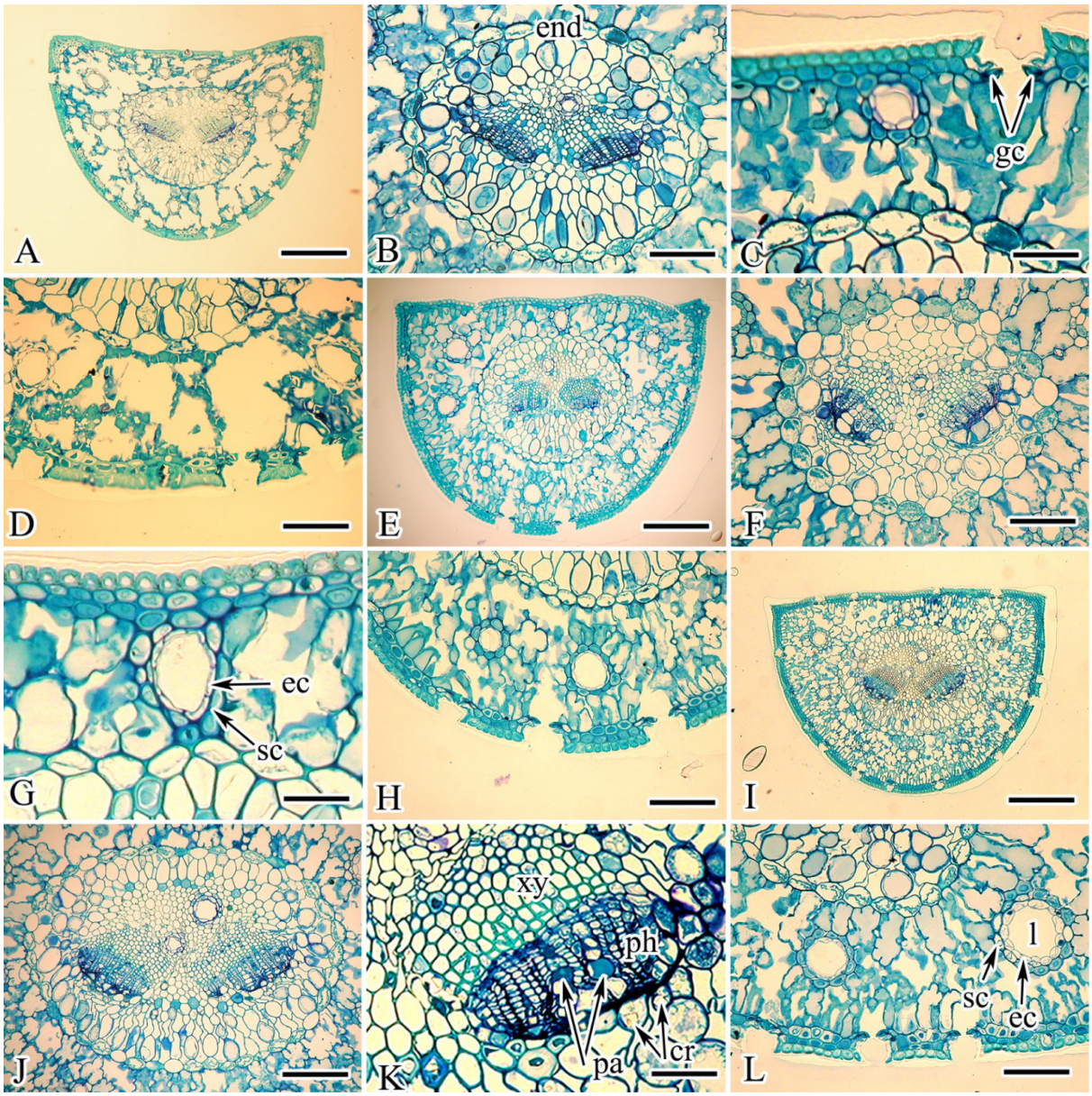


Fig. 4. Cross section (CS) of *Pinus thunbergii* leaf. **A-D.** Samples of Taean. **A.** CS of whole leaf. **B.** Vascular bundle. **C.** Adaxial epidermis showing stomata, and external resin duct. **D.** Abaxial epidermis. **E-G.** Samples of Ullung-do. **E.** Whole leaf. **F.** Vascular bundle. **G.** Adaxial epidermis and external resin duct. **H.** Abaxial epidermis. **I-L.** Samples of Jeju-do. **I.** Whole leaf. **J.** Vascular bundle. **K.** Single vascular bundle showing crystals and parenchyma bands. **L.** Abaxial epidermis and resin duct. Abbreviations: cr, crystals; ec, epithelial cells; gc, guard cells; pa, parenchyma bands; ph, phloem; rd, resin duct; sc, sheath cells; vbs, vascular bundle sheath; xy, xylem. Scale bars: A, E, I = 75 μm; B, C, D, F, G, H, J, L = 20 μm; K = 10 μm.

Results

Pinus thunbergii has two needle leaves with about 12 to 18 cm long. The apex of needles has pointed tip. In cross section, needles are semicircular in outline. The epidermis covered with an extremely thick cuticle. The highly sunken stomata are arranged in rows and found throughout the leaf surface. Detailed description and general arrangement of tissues is given below.

Epidermis

The cross section of the needle reveals a typical unifacial structure (Fig. 3, 4). The adaxial side is flat and abaxial side is semi-circular (Fig. 3A, E, H, L; 4A, E, I). An epidermis is unistratified and composed of round or polygonal fibrous cells and covered by cuticle (Fig. 3C, G, J, N; 4C, D, H, L). The continuity of the epidermis is broken by the presence of numerous stomata with small substomatal cavities (Fig. 3C, G,

Table 1. The position of resin ducts in *Pinus thunbergii* leaf.

Location	Site of collected population	No. of vascular bundle	No. of resin ducts			No. of stoma rows in abaxial side
			contact with hypodermis	in mesophyll	between the vascular bundle and vascular bundle	
Goseong (Gangwon)	1	2	1	9	2	8
Gangneung	1	2	0	6	0	8
Uljin	1	2	0	4	0	8
Pohang	1	2	0	8	1	8
Busan	1	2	0	10	2	7
Tongyoung	1	2	1	6	1	7
	2	2	0	4	1	6
	3	2	0	2	1	6
Weinarodo	1	2	0	5	1	8
	2	2	0	2	1	6
	3	2	0	5	1	7
Jindo	1	2	0	4	2	5
	2	2	0	5	1	7
	3	2	0	5	1	7
Gochang-gusipo	1	2	0	3	1	6
	2	2	0	7	1	7
	3	2	0	6	1	9
Taean	1	2	1	2	1	4
	2	2	1	5	2	6
	3	2	0	7	2	7
Ullung-do	1	2	1	6	1	5
	2	2	1	6	2	7
	3	2	0	3	1	5
Jeju-do	1	2	0	4	1	7
	2	2	0	9	1	7
	3	2	0	8	2	7

J, N). No epidermal dissimilarity observed in all samples collected from different sites.

Hypodermis

The single layered epidermis followed by 2-3 layered hypodermis. Hypodermal cells are with strongly thickened wall and sclerenchymatous in nature. In most part hypodermis is one to two layered; only in few parts it is three layered and even more in angles. The shape and size of hypodermal cells are highly variable (Figs. 3C, D, G, J, N, 4C, D, G, H, L). Stomata also broke the continuity of hypodermis; in fact the guard cells arise from hypodermis (Figs. 3C, 4C). In the leaf samples of Goseong, Taean and Ullung-do an external resin canal also associated with hypodermal region (Figs. 3D, 4C, G; Table 1).

Mesophyll

The mesophyll is monomorphic and comprised of thin-walled cells containing numerous chloroplasts and abundant starch. The cells are peculiar in shape: cell walls have numerous small infolding and project into the cavities of the cells, a characteristic cell type of *Pinus*. The mesophyll occupy the most part almost two third of leaf area and contains a number of resin ducts. The total number of resin ducts in *P. thunbergii* studied here ranges from 4 to 12. The highest number of resin

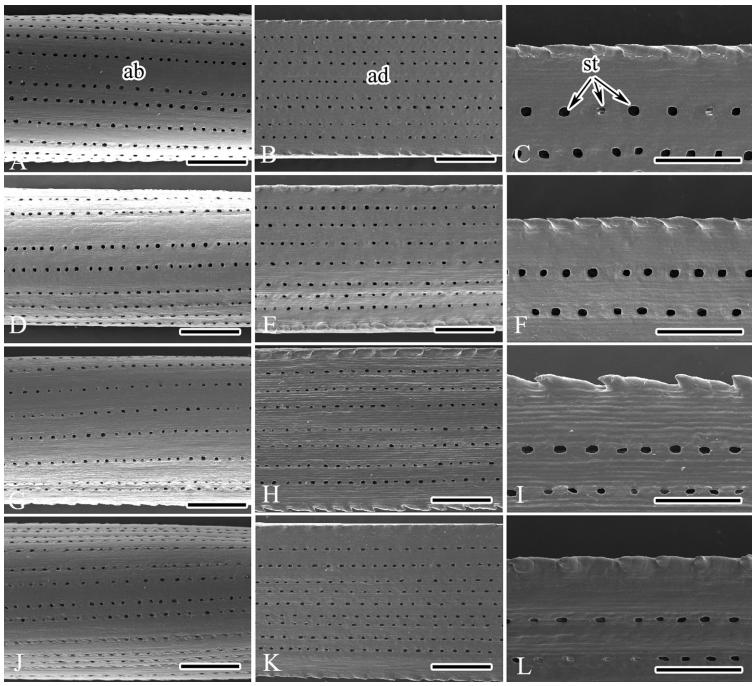


Fig. 5. SEM micrograph of *Pinus thunbergii*. **A-C.** Samples of Gangneung. **A.** Abaxial surface. **B.** Abaxial surface. **C.** Serrated margin of leaf. **D-F.** Samples of Tongyoung. **D.** Abaxial surface. **E.** Abaxial surface. **F.** Serrated margin of leaf. **G-I.** Samples of Weinarodo. **G.** Abaxial surface. **H.** Abaxial surface. **I.** Serrated margin of leaf. **J-L.** Samples from Gochang-gusipo. **J.** Abaxial surface. **K.** Abaxial surface. **L.** Serrated margin of leaf. Abbreviations: ab, abaxial surface; ad, adaxial surface; st, stomata. Scale bars: A, B, D, F, G, H, J, K = 500 μm; C, F, I, L = 300 μm.

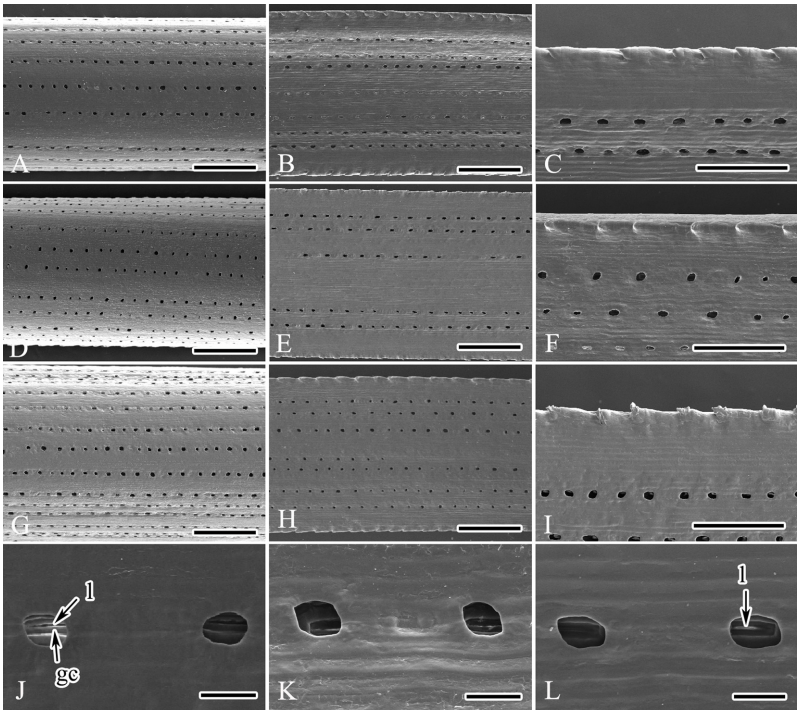


Fig. 6. SEM micrograph of *Pinus thunbergii*. **A-C.** Samples of Taean. **A.** Abaxial surface. **B.** Abaxial surface. **C.** Serrated margin of leaf. **D-F.** Samples of Ullung-do. **D.** Abaxial surface. **E.** Abaxial surface. **F.** Serrated margin of leaf. **G-I.** Samples of Jeju-do. **G.** Abaxial surface. **H.** Abaxial surface. **I.** Serrated margin of leaf. **J-L.** Stomata in detailed. **J.** Tongyoung. **K.** Ullung-do. **L.** Jeju-do. Abbreviations: gc, guard cell; l, stomata lumen. Scale bars: A, B, D, E, G, H = 500 μm; C, F, I = 300 μm; J, K, L = 100 μm.

ducts found in Goseong and Busan, whereas the lowest found in Uljin and Weinarodo (Table 1). Two types of cells constitute the resin duct: a thick-walled sheath cells covering a layer of thin-walled epithelial cells (Fig. 4G, H, L). Number of sheath and epithelial cells depend on the size of duct, a small duct has few cells and a large one has many cells.

Vascular bundle

The central tissue of leaf is enclosed by well represented single layered endodermis with Casparian strips. There are two vascular bundles within endodermis which are subdivided by several layered parenchymatous band (Figs. 3B, F, I, M, 4B, F, J, K). In the vascular bundle xylem places to the adaxial side and phloem to abaxial side. In all samples a layer or two of fiber cells just below the phloem are found. Some parenchyma cells around the bundle are filled with crystals (Figs. 3K, O, 4K). The crystals also found in some phloem parenchyma cells. With the exception of the samples from Gangneung and Uljin there are one or two resin ducts inside bundle sheath (Table 1) and they are usually places towards xylem side.

Stomata

Pinus thunbergii leaves are amphistomatous, stomatal pores found on both sides. Stomata are arranged in stomatal rows on both abaxial and adaxial sides of leaf (Figs. 5, 6). The numbers of such rows are high in abaxial side though it covers large area of the leaf itself than flat adaxial side (Figs. 5A, D, G, J, 6A, D, G). Leaf samples from different locations mainly disagree each other in the number of stomatal rows in flat adaxial side and number varies from six in Ullung-do to nine in Gochang-gusipo (Figs. 5B, E, H, K, 6B, E, H; Table 1). The leaf surface is smooth, no external partition or decoration on subsidiary and epidermal cells observed in SEM study. However, two edges or margin of the leaf are serrated, small spine like structures developed on the margin (Figs. 5C, F, I, L, 6C, F, I). The spine like outgrowth are found throughout the length and pointed towards the tip of the leaf. Stomata are sunken type. The guard cells are deeply seated and arise from the hypodermal region of the leaf (Figs. 4C, H, L; Figs. 6J, K, L).

Discussion

We found the needle of *P. thunbergii* are supplied with two vascular bundles surrounded by common bundle sheath. However, this is not a common case in *Pinus* as some species have one vascular bundle surrounded by bundle sheath like *P. monophylla*, and *P. wallichinana* (Dorken and Stutzel, 2012). Except *P. thunbergii* the species with two vascular bundles within common bundle sheath studied so far are *P. resinosa* (Gambles and Dengler, 1982) *P. taiwanensis* (Sheue et al., 2003) *P. sylvestris* (Dorken and Stutzel, 2012), and *P. roxburghii* (Tiwari et al., 2013). In fact, this is quite uncommon feature on the plant kingdom and the functional meaning as well as the evolutionary pathway leading to the two bundles within single bundle sheath remains thus unclear (Dorken and Stutzel, 2012). However, some of the previous studies explain that two versus one single vascular bundle in *Pinus* has been proven to be a useful diagnostic feature, for example difference in subgenus *Strobus* and *Pinus* (Eckenwalder, 2009; Tylor et al., 2009).

The basic anatomical features of *P. thunbergii* leaf are typical pine needle type with sunken stomata throughout the leaf surface; prominent cuticle covers the epidermis, hypodermis with strongly thickened walls, monomorphic mesophyll with strange cells type and number of resin ducts, distinguished bundle sheath covering two vascular bundle. One/two resin ducts also found inside bundle sheath in some samples. One interesting structure found in *P. thunbergii* is presence of parenchymatous bands within the xylem and phloem stands. Although number of such bands is vary in different samples but in thickness all are one cell row broad. In their study on *P. monophylla*, *P. wallichiana*, and *P. sylvestris* Dorken and Stutzel (2012) describe this structure only in *P. monophylla* and *P. wallichiana*. Interestingly enough, *P. thunbergii* shows the duel character here because it resembles with *P. sylvestris* in number of vascular bundle since these both have two vascular bundles instead of one in *P. monopbylaa* and *P. wallichiana*. Some parenchyma cells around the vascular bundle filled probably with crystals of calcium oxalate. In some samples these crystals appear in parenchyma bands within vascular bundle too. Intracellular calcium oxalate crystals within the ray cells of needle phloem have been reported in *P. palustris* (Pritchard et al., 1997). The presence/formation of such crystals in the needle tissue probably has a great adaptive value in xeric environment.

In this study, we found resin duct is the most inconsistent feature within the species. The number of resin duct varies from 4 to 12 in *P. thunbergii* from different coastal regions of Korea. In general, the resin ducts has been considered as an important character and applied in classifying the Pinaceae and particularly in distinguishing *Pinus* species (Boratynska and Bobowicz, 2001; Fu et al., 1999; Richardson, 1998) and even used to differentiate hybrids (Kormutak et al., 1993). However, from this study it is difficult to agree with this fact because even same species from different location vary in number and position of the resin ducts. It is clear that the variation found in *P. thunbergii* in this study is probably related to environmental factors, which is already described in some previous studies (Hengxiao et al., 1999; Jokela et al., 1998; Sheue et al., 2003; Skuterud et al.,

1994; Telewski et al., 1999 Telewski et al., 1983). The southern region of Korea, particularly Busan, is warmer 2°C than Ullung-do and other regions (Fig. 2A). Correspondingly, precipitation is also relatively higher in southern part than east and west regions (Fig. 2B). We found that samples from Goseong, Taean and Ullung-do have single external resin ducts and samples from Gangneung and Uljin have no resin ducts inside bundle sheath. However, the external and duct inside bundle sheath are not much taxonomically important. It is generally accepted that only the resin ducts in the mesophyll are useful for species identification since the presence of resin ducts in other region are not consistent (Sheue et al., 2003).

Pinus leaves are amphistomatous, stomatal pores found on all sides. In *P. thunbergii* stomata are arranged in stomatal rows on both semicircle abaxial and flat adaxial side. Although, it is difficult to count the number of stomata row in abaxial side but has obviously more than adaxial side because it covers large area of the leaf. The number of rows in stomatal bands has great taxonomic value in the Taxad genera and particular genera/species represented by same range of stomata rows (Spjut, 2007). However, studies with respect to exact number of stomata rows and its taxonomic value in *Pinus* species are still lacking in literatures. The problem here in *Pinus* is: the species have no particular shape with clear external abaxial and adaxial differentiation. In many species of *Pinus* the only way to know abaxial and adaxial side is the position of xylem and phloem. Here in this study, sample from Gochang-gusipo has nine stomatal rows in fact maximum among the studied samples and Ullung-do has six the minimum number. Gochang-gusipo has medium precipitation and temperature among the five sites and Ullung-do has slightly high precipitation than Gochang-gusipo but lowest temperature among the sites. By this study, we presumed that the variation in stomata rows on adaxial surface not essentially linked with temperature and precipitation because Taean in west coast along with Gochang-gusipo has seven stomata rows similar with Jeju-do and Southern coast.

In conclusion, we observed some variations on *P. thunbergii* leaf traits from different coastal regions of Korea. The dissimilarities basically found in number and position of resin ducts and number of stomata row in leaf surface. Besides several environmental factors, insect attacks and phytohormones are affected to variation in the distribution and number of resin ducts in the plant body (Helmers, 1943; Fahn and Bemayoun, 1976). On the other hand stomatal arrangements correspond to light intensity, an increase in light intensity results in an increase stomatal index. Thus, we conclude that the difference in climatic conditions of dissimilar coastal region in Korea must be responsible for these leaf trait variations. However, further morphological and genetic studies of the species are highly recommended to verify these inconsistencies.

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